Why do people smoke ?

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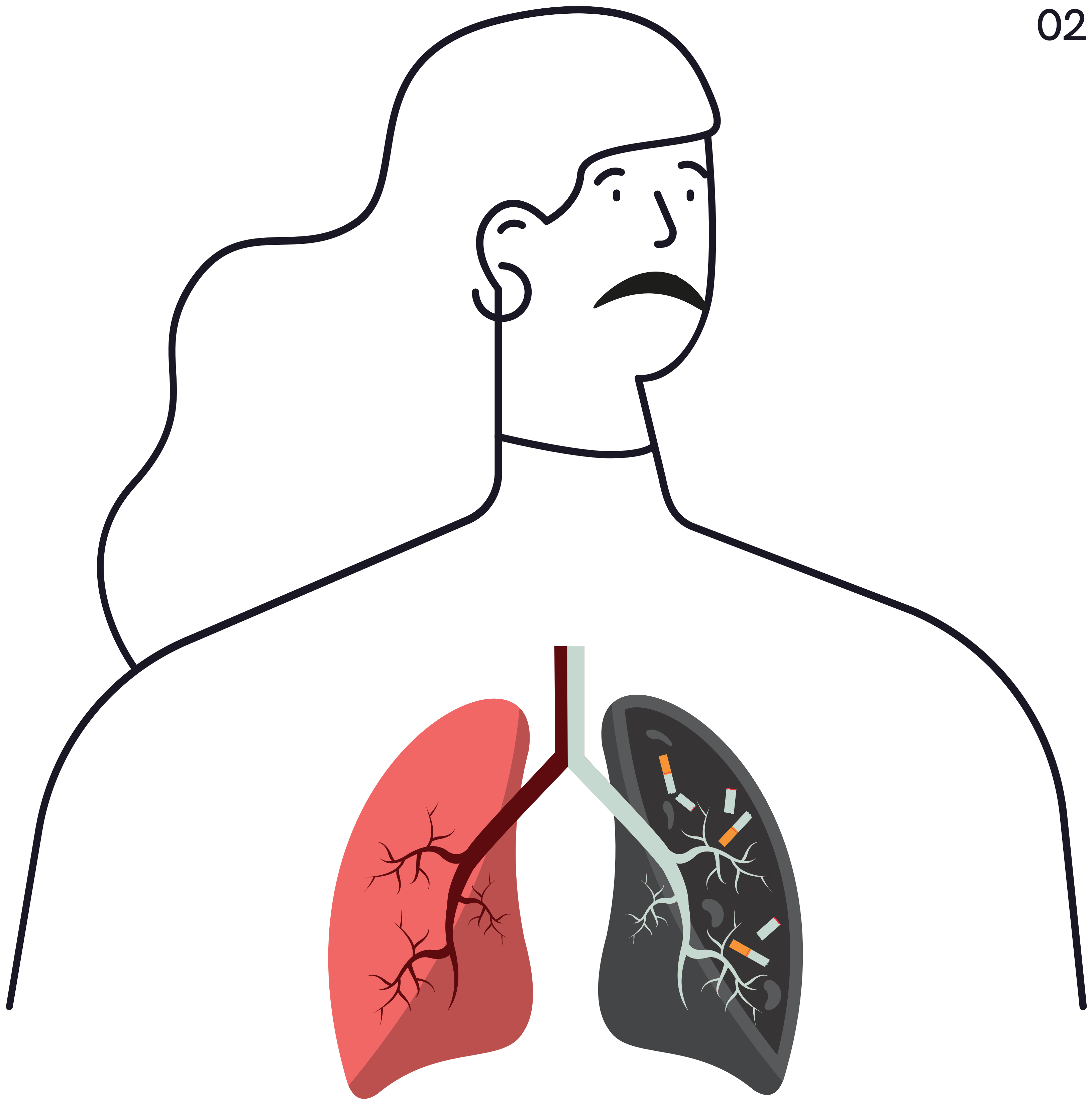
Springboard

Why do we care ?

IN 2023,
GLOBALLY ~7 MILLION LIVES WERE LOST
USA ~500,000 LIVES

POLICY INITIATIVES ARE EVERYWHERE , BUT
ARE THEY REALLY **WORKING ??**



What we want to answer ?

What factors (defined by Health Belief Model) (HBM) contribute most towards the habit of smoking ? Are there any policy implications?

Does this trend vary by age group ?

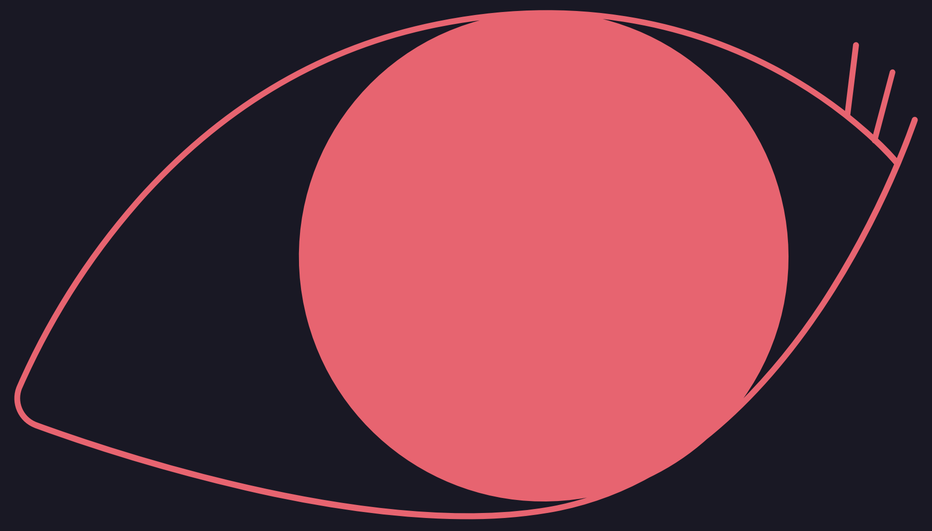
DATA

DATASET: BEHAVIORAL RISK FACTOR SURVEILLANCE SYSTEM (BRFSS)

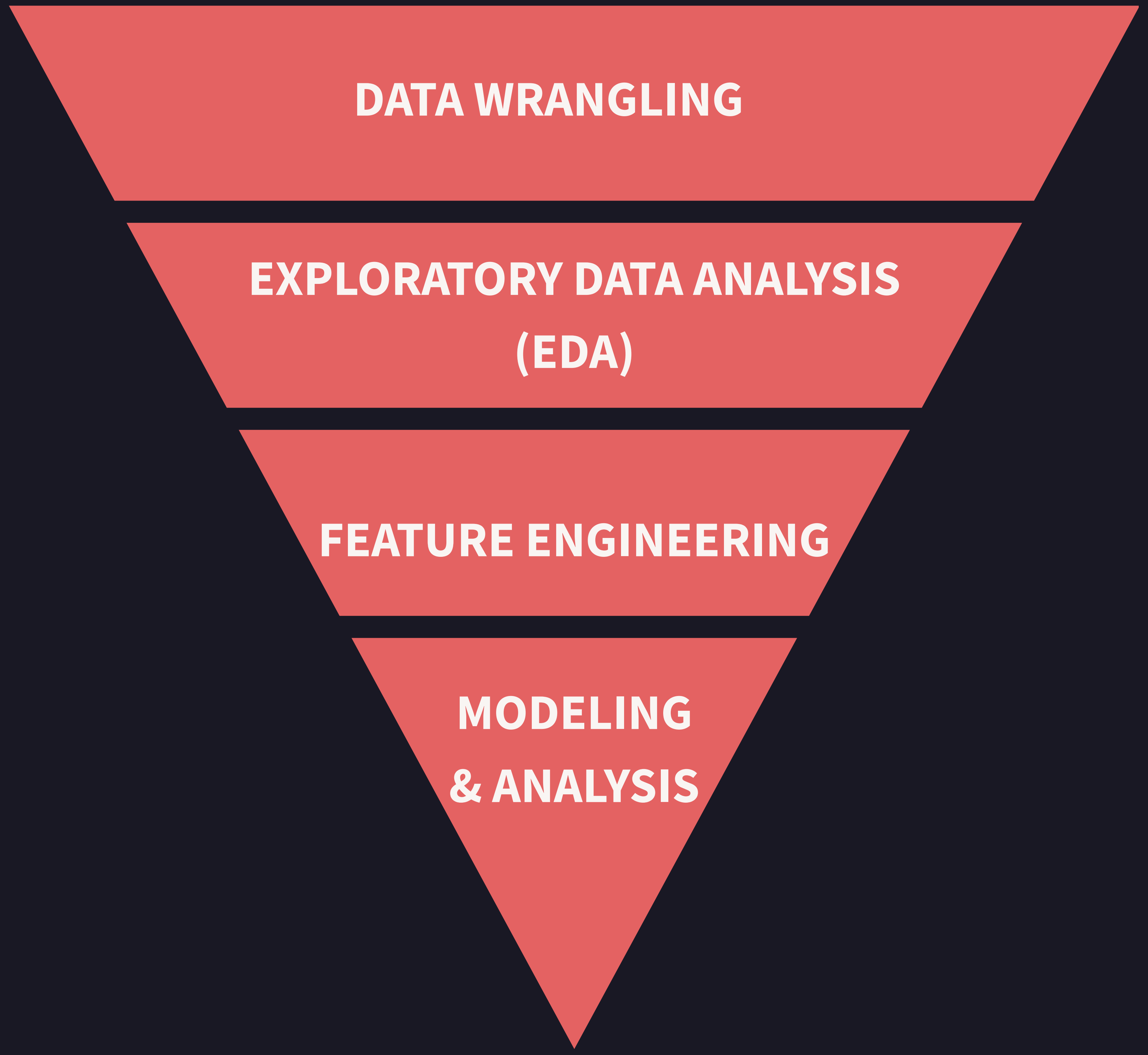
SOURCE: KAGGLE API

SIZE: ~ 2MILLION RECORDS AND 645 COLUMNS

DATA TYPE: SURVEY



METHODOLOGY



SMOKER PREDICTION MODEL

HEALTH BELIEF MODEL

Widely used frameworks in public health (since 1950s) for understanding why people do or don’t take certain health action

Perceived Susceptibility
Belief about how likely you are to get a disease or condition.

Perceived Severity
Belief about how serious the consequences would be if the disease occurs.

Perceived Benefits
Belief that taking a specific action will reduce the risk or severity of the condition.

Other Variables
Demographic, sociopsychological, and structural variables affect an individual's perceptions of quitting smoking.

Perceived Barriers
Belief about the costs, challenges, or obstacles to taking the action.

Cues To Action
Events, people, or things that prompt the decision to act.

Self - Efficacy
Confidence in your ability to successfully take the action.

DATA WRANGLING

MISSING DATA

Drop columns with missingness > 0.75

```
threshold = len(combined_df)*.75
df.dropna(axis=1, thresh=threshold)
```

645 to 87 columns

NON-INFORMATIVE FEATURES

Drop redundant features

```
df.drop(['SEQNO', 'CTELENUM'], axis=1)
```

87 to 84 columns

DUPLICATE FEATURES

Drop record identifiers

```
df.drop(['INCOME2'], axis=1)
```

84 to 58 columns

WRONG DATATYPES

Change datatypes appropriately

_EDUCAG and _INCOME Ordinal variables

PHYSHLTH and MENTLHLTH Numeric variables

No Change

HANDLING ERROR VALUES & IMPUTE FEATURE VALUES

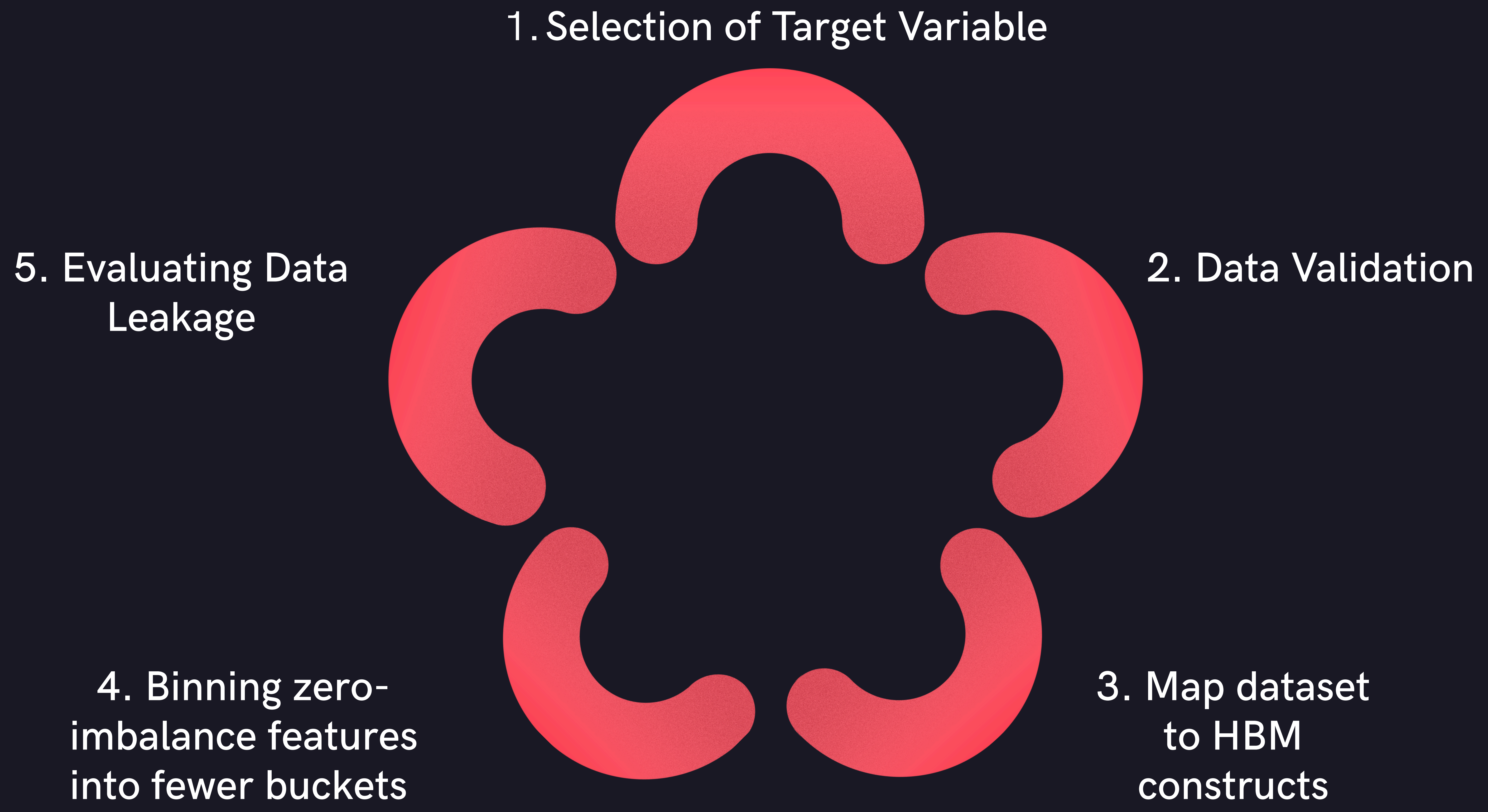
Standardize feature values to same units

Drop _IYEAR, _IMONTH, _IDATE

Impute _HEIGHT2, _WEIGHT3 with mean

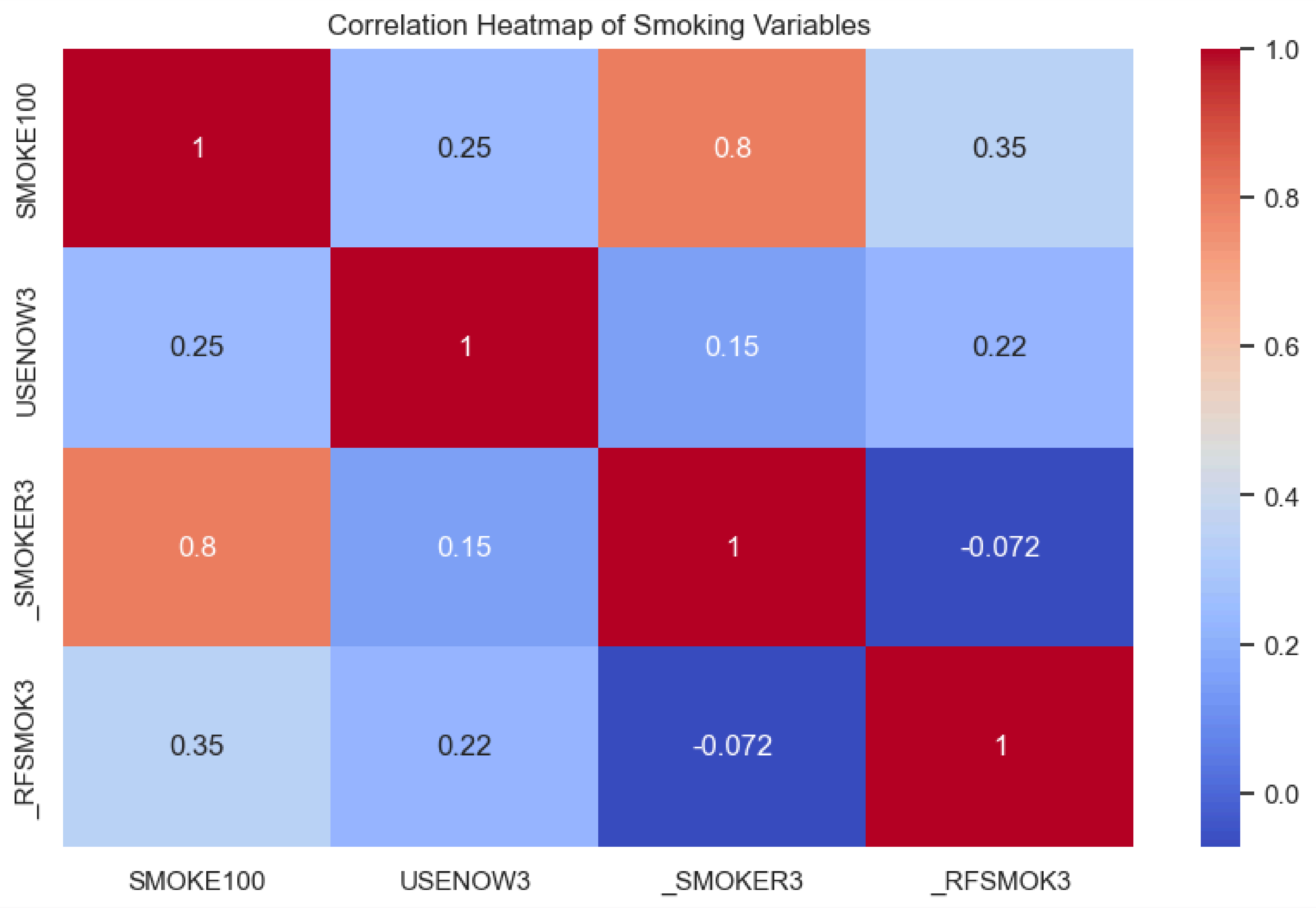
58 to 55 columns

EDA


1. Selection of Target Variable
2. Data Validation
3. Map dataset to HBM constructs
4. Binning zero-imbalance features into fewer buckets
5. Evaluating Data Leakage

Selection of Target Variable

TARGET VARIABLE	DEFINITION	DISTINCT CLASSES
SMOKE100	Smoked at Least 100 Cigarettes	4 (Yes, No, Don't know, Refused)
USENOW3	Use of Smokeless Tobacco Products	5 (Every day, Some days Not at all, Don't know,Refused)
_SMOKER3	Computed Smoking Status	5 (Everyday smoker, Someday smoker, Former smoker, Non-smoker)
_RFSMOK3	Current Smoking Calculated Variable	3 (No, Yes, Don't know)



Map dataset to HBM constructs

Variable	HBM Construct(s)	Explanation
ASTHMA3	Susceptibility	Having asthma could indicate if the person is likely to get risks related to smoking
PHYSHLTH	Severity	number of days of poor physical health indicates how serious the risks of smoking are for the person
HLTHPLN1	Benefits	Having a health insurance helps reassure that the risks related to smoking is reduced
MEDCOST	Barriers	Medical cost; barrier to access care/cessation support which could affect how one cares for ones health ie. might increase smoking
CHECKUP1	Cues to Action	Routine checkup; triggers cues and reinforces benefits.
EXERANY2	Self- Efficacy	If the person has a self motivation to stay healthy by prioritising exercise, he has better motivation to continue taking healthy actions.
MARITAL	Other Variables	Socioeconomic factor that determines certain behaviour

Evaluating Data Leakage

Chi-square test: To check linear relationship between target variable and each feature. Cramer's V evaluates strength of relationship

Observation: *Socioeconomic factors (education, income, marital status) have more straightforward linear relationships*

Mutual Information test: Measures non-linear relationships and information content shared between variables

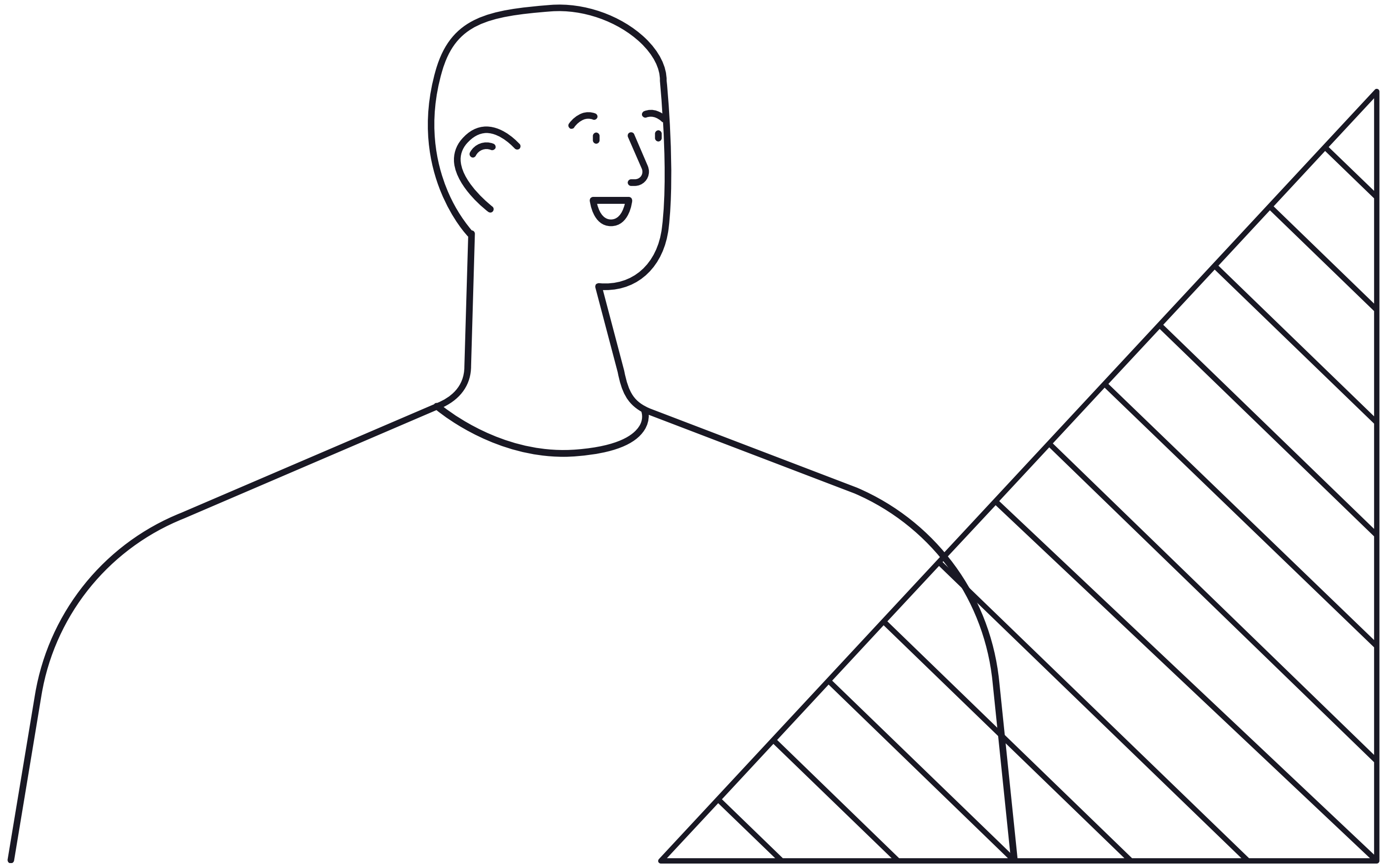
Observation: *Medical conditions (stroke, kidney disease, heart attack) have complex, non-linear relationships with your target that Chi-Square missed.*

FEATURE ENGINEERING

CATEGORICAL VARIABLES:
One-hot Encode

NUMERIC VARIABLES:
Standardize

GENERATE TRAIN/TEST SPLIT:
~1.9 million records for Train set
~500,000 records for Test set



MODELING

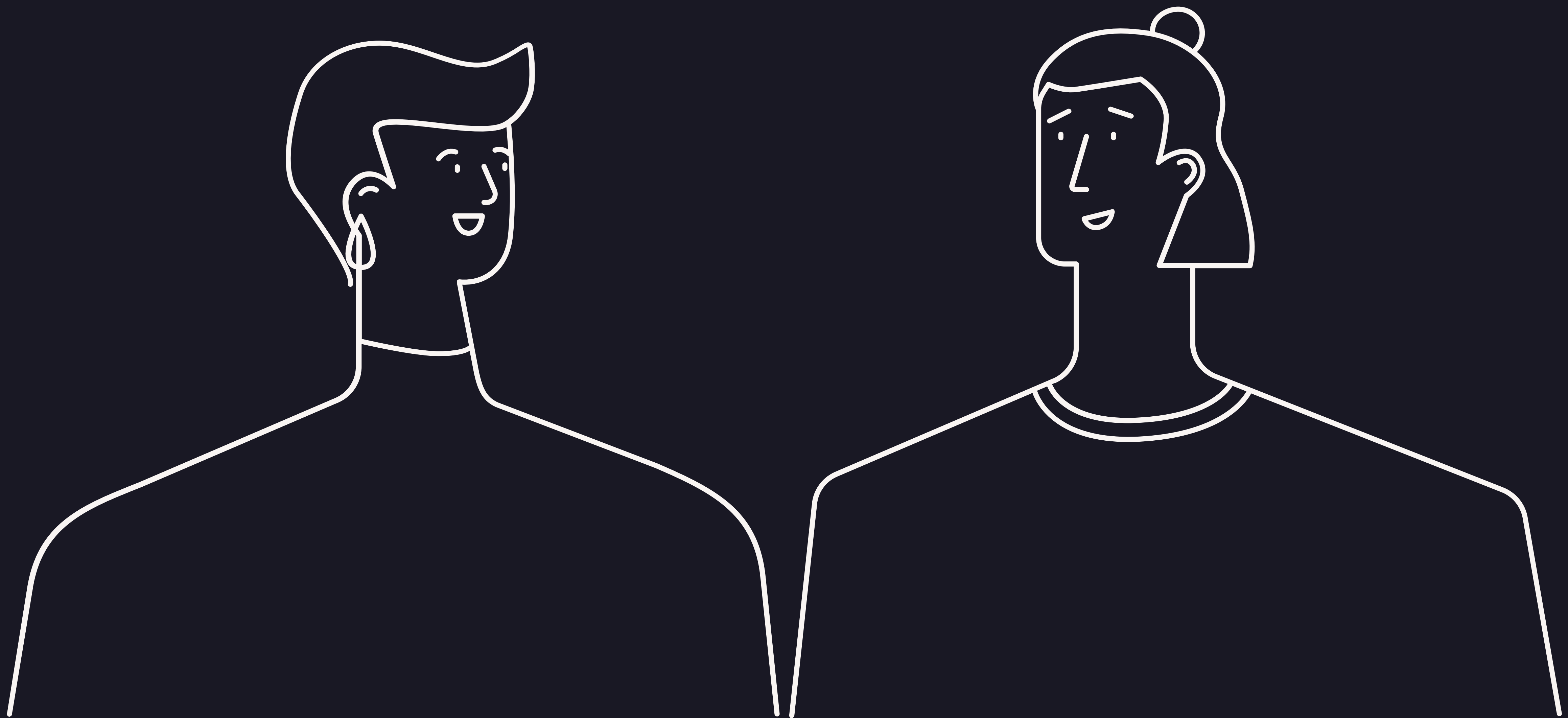
Model	Setup	Imbalance Handling	Key Params	Early Stopping
CatBoost Default	Minimal, fast baseline	auto_class_weights='Balanced'	Defaults	❌
CatBoost Tuned	Optimized for imbalance	Same as default	iterations=1000, depth=8, learning_rate=0.1, regularization params	(100)
LightGBM Default	Basic config	class_weight='balanced'	num_leaves=31, lr=0.1, n_estimators=1000	50
LightGBM Tuned	RandomizedSearchCV tuned	Same as default	Tuned (num_leaves, depth, learningrate, reg_alpha/lambda)	30

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Training Time
CatBoost	0.729721	0.333598	0.713685 [1]	0.454670 [2]	0.796675 [2]	0.458032 [2]	1144.751788
CatBoost Tuned	0.730858	0.334517	0.712342	0.455249	0.796907	0.458349	1831.023640
LightGBM	0.728925	0.332754	0.713301	0.453807	0.795863	0.456280	16.839819
LightGBM Tuned	0.730335	0.333869	0.711520	0.454481	0.796365	0.457688	3298.030455

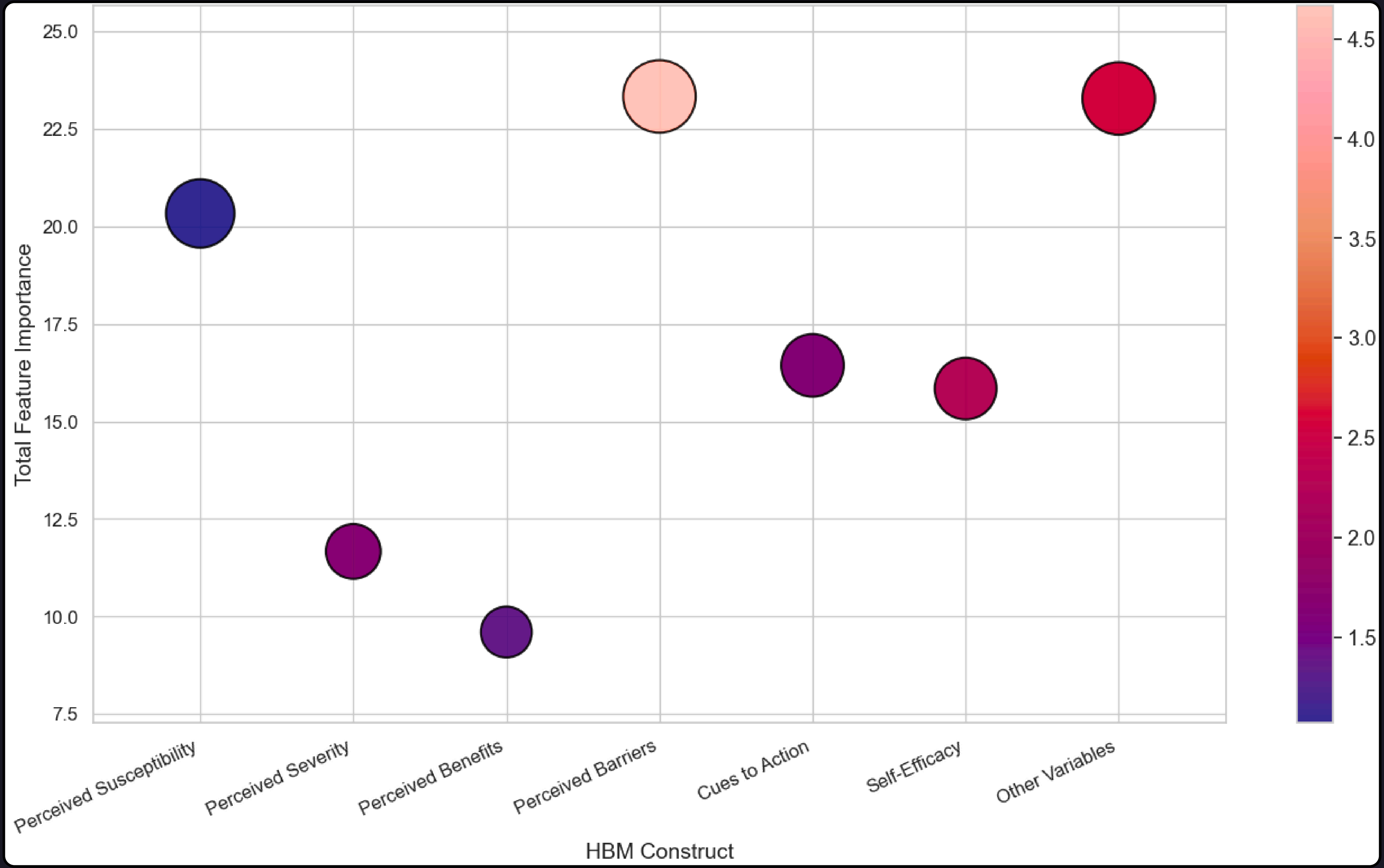
ANALYSIS

What constructs defined by Health Belief Model (HBM) contribute most towards the habit of smoking ? What are some policy implications ?

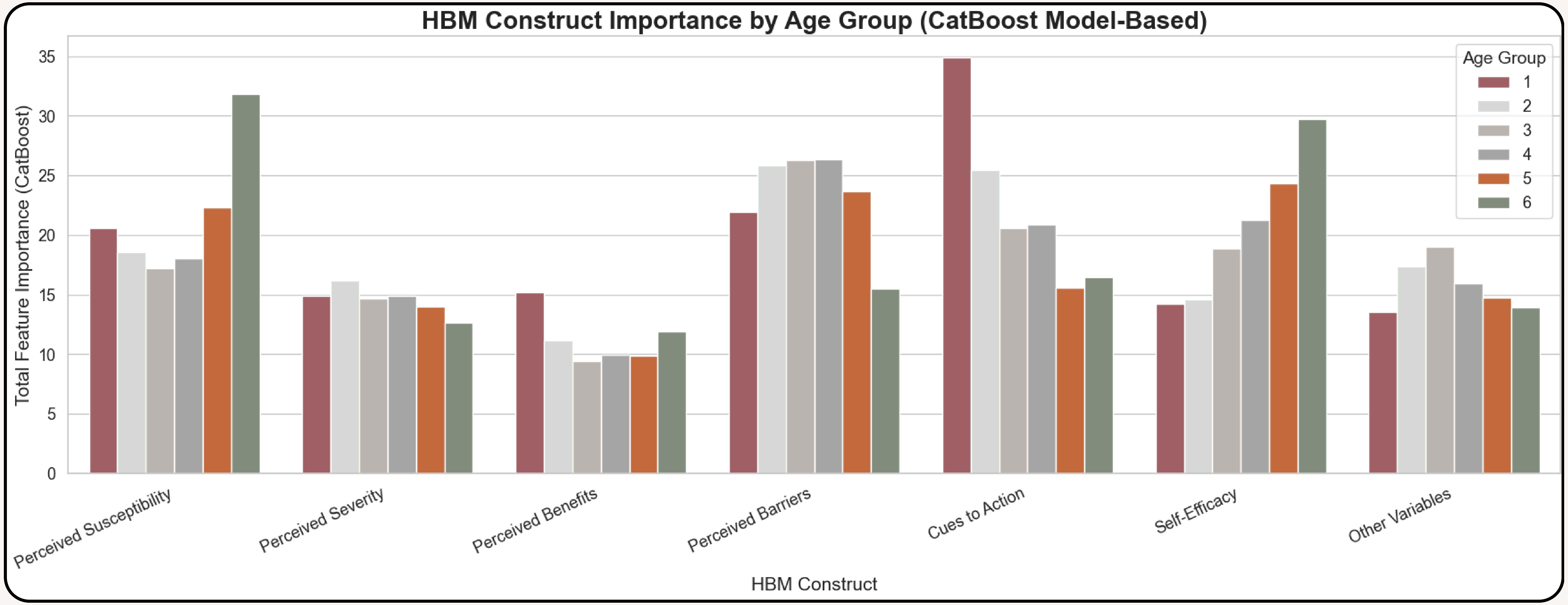
Does this trend vary by age group ?



TOP PREDICTORS



Feature	Importance (Prediction ValuesChange)	HBM Construct(s)
_EDUCAG	15.54	Barriers
_AGE65YR	9.62	Other
MARITAL	7.18	Other
_RFBING5	6.01	Cues to Action
WEIGHT2	5.06	Self-Efficacy
CHCCOPD1	4.89	Susceptibility
GENHLTH	4.8	Severity
_INCOMG	4.16	Barriers
HEIGHT3	3.47	Self-Efficacy



Code	Age Range	Description
1	18–24	Young Adults
2	25–34	Early Career Adults
3	35–44	Mid-Career Adults
4	45–54	Mature Adults
5	55–64	Pre-Retirees
6	65+	Seniors

18-24

Most influenced by **Cues to Action** and Perceived Benefits

Leverage peer influence, social media, benefits messaging.

25-54

Dominated by **Barriers**.

Address barriers: stress, time, cost, support access.

55-64

Influenced most by **Susceptibility**.

Emphasize risk and susceptibility via health condition links.

65+

High on **Self-Efficacy**, **Susceptibility**, and **Benefits**.

Empower with tools and reinforce confidence to quit.

Future work

I would also like to add more recent data to my project (~2021-2024) to evaluate differences in smoking habits and differences in constructs over a decade.

Include some local public health officials to gather their feedback.

Possibly publish a research paper in the AJPH (American Journal of Public Health)